

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison



Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	SHAKTHI M
Register Number	192511197
Class / Slot	Slot C
Capstone Title	SleepScope: Student Sleep and Academic Stress Analytics App
Capstone Product Type	Web app / Android app / iOS app / Hybrid app

CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available
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Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	College students
User roles	Student, Admin
Main modules	User Authentication, Sleep Tracking, Stress Assessment, Analytics Dashboard, Notifications, Reports
Cloud services identified in CO1	Cloud Virtual Machine, Cloud Storage, Managed Database, Analytics Service
API/backend need	REST APIs for login, sleep/stress data submission, analytics retrieval
Database need	Store user profiles, sleep records, stress questionnaires, analytics results
File upload/storage need	Optional upload of sleep reports or health documents
Analytics/AI/Python module	Python-based analytics to identify sleep trends and academic stress levels

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: Analytics App with AI/Python Module

The application performs user authentication, stores daily sleep records, analyzes stress patterns, generates dashboards, and predicts stress levels using Python analytics. Therefore, it requires more processing than a simple CRUD application.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Concurrent Active Users	Registered Users x Active User % x Peak Factor	$500 \times 10\% \times 2 = 100$
Requests per Minute	Concurrent Active Users x Actions per User per Minute	$100 \times 2 = 200$
Requests per Second	Requests per Minute / 60	$200 / 60 = 3.33 \text{ RPS}$
Daily Requests	Daily Active Users x Avg Actions per Day	$200 \times 30 = 6000$
Input	Value	Reason / Assumption
Registered users expected	1000	The app is intended for use by students in one or two colleges during the initial deployment.
Active user percentage	20%	Around 20% of registered users are expected to use the app daily.
Peak factor	2	User activity is expected to double during peak hours (morning and evening).
Actions per user per minute	3	A user may log sleep, view analytics, and check stress reports within one minute.
Calculated RPS	20 Requests/Second (RPS)	Concurrent Users = $1000 \times 20\% \times 2 = 400$. Requests/Minute = $400 \times 3 = 1200$. RPS = $1200 \div 60 = 20 \text{ RPS}$.
Daily active users	200	Approximately 20% of registered users are active each day.
Daily requests	6000	Daily Active Users = 200. Average Actions per Day = 30. Therefore, $200 \times 30 = 6,000 \text{ requests/day}$.

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Base web/API application	1
Authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2
Background jobs or notifications	+1
Analytics or dashboards	+2
AI/Python inference inside server	+2 to +4
Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU
5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service

10+		Scale out with multiple VMs or managed compute	
CPU Driver Selected		Score	Reason
Base web/API application	1	Standard mobile backend	
Authentication and roles	+1	Student/Admin login	
API traffic	+1	Around 20 RPS	
Database/search/reporting	+1	Frequent data retrieval and reports	
Background jobs	+1	Daily reminders and notifications	
Analytics/dashboard	+2	Dashboard generation	
AI/Python module	+2	Sleep and stress prediction	
Total CPU Score	9		
Recommended vCPU	4-8 vCPU		

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula

Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2 GB
Web/API service	1-2 GB	2 GB
Small database	2-4 GB	4 GB
Medium database/cache	4-8 GB	4 GB
Analytics/reporting	4-8 GB extra	4 GB
AI/Python module	4-16 GB extra	16 GB
25% buffer	Subtotal x 0.25	4 GB
Final RAM	Round up to standard size	20 GB TO 24 GB

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
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OS and application	20-40 GB	30 GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	20 GB
File uploads	Avg file MB x files/day x days x 1.3	15 GB
Logs	Requests/day x log KB x days / 1024 MB	5 GB
Backup/growth buffer	Subtotal x 30%	21 GB
Final storage	Round up to standard disk size	100 GB SSD

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	4 vCPU
Two-tier design	App VM + managed/separate database	24 GB
Three-tier design	Frontend, backend/API and database separated	100 GB SSD
Containerized backend	Microservices, portable API, CI/CD	Managed Cloud Databases
Serverless function	Event-based task, notification, small automation	100
Managed database/storage	Reliability, backup and scaling required	40 GB

Final VM Plan: Two-tier Design

Separating the application server from the managed database improves security, scalability, and maintenance while keeping deployment simple for a student prototype.

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments
Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment
Context	Selected Approach	Justification

Local prototype/testing	Type 2 Hypervisor (VirtualBox)	Easy local development
Cloud pilot deployment	Cloud-managed Virtualization	Managed infrastructure with minimal administration
Production growth scenario	Type 1 Hypervisor	High performance and reliability
Module suitable for container	Backend API	Easy deployment and scalability

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Every release	Rollback if deployment fails
Before database schema change	Snapshot	Before every update	Protect database
Before OS/package update	Snapshot	Monthly	Restore system if update fails
Before review/demo	Snapshot	Before presentation	Stable demonstration
Create testing environment	Clone	Every sprint	Independent testing
Create team demo copy	Clone	When needed	Separate demo environment

Technical Tip

A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	VM / Container / Managed Service	Reason
Frontend	VM	Stable hosting
Backend/API	Container	Easy scaling and deployment
Database	Managed Service	Automated backup and maintenance
File storage	Managed Object Storage	Scalable and secure
AI/Python module	Container	Independent resource allocation

Analytics/dashboard	Container	Flexible deployment
Notification/background jobs	Container	Lightweight background processing

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

SleepScope is best deployed using a two-tier cloud architecture consisting of an application VM and a managed cloud database. The application server should be configured with 4 vCPUs, 24 GB RAM, and 100 GB SSD storage to support analytics, dashboards, and AI processing. Backend services and AI modules should run in containers for portability and scalability, while the database should be hosted as a managed cloud service. Cloud-managed virtualization is recommended for deployment, with Type 2 virtualization used during local development and Type 1 hypervisors supporting production environments. Snapshots should be created before deployments, database updates, and operating system upgrades to enable fast recovery. Clones should be used for testing and demonstration environments. Object storage should be used for uploaded files instead of VM disks. This infrastructure provides sufficient performance, scalability, reliability, and maintainability for the SleepScope application prototype while allowing future growth.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.